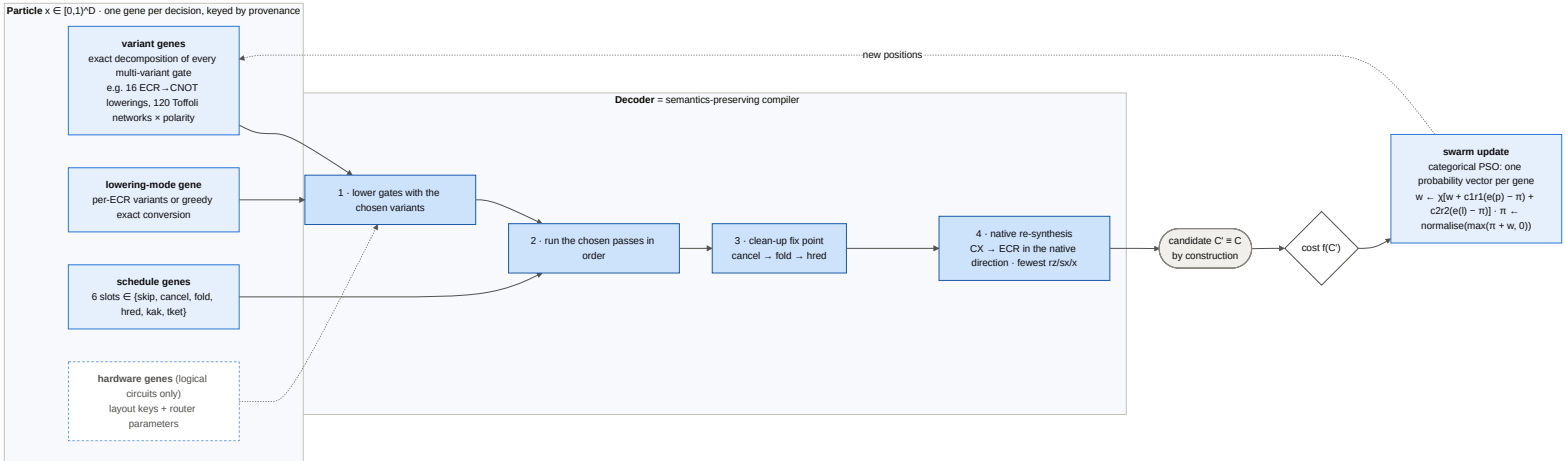

Particle x ∈ [0,1)^D · one gene per decision, keyed by provenance
variant genes
exact decomposition of every multi-variant gate
e.g. 16 ECR → CNOT lowerings, 120 Toffoli networks × polarity
lowering-mode gene
per-ECR variants or greedy exact conversion
schedule genes
6 slots ∈ {skip, cancel, fold, hred, kak, tket}
hardware genes (logical circuits only)
layout keys + router parameters
Decoder = semantics-preserving compiler
1 · lower gates with the chosen variants
2 · run the chosen passes in order
3 · clean-up fix point
cancel → fold → hred
4 · native re-synthesis
CX → ECR in the native direction · fewest rz/sx/x
candidate C′ ≡ C by construction
cost f(C′)
swarm update
categorical PSO: one probability vector per gene
w ← χ[w + c1r1(e(p) − π) + c2r2(e(l) − π)] · π ← normalise(max(π + w, 0))
new positions